

Regresión y correlación

$$\hat{y} = b_0 + b_1 x \quad b_1 = r \frac{s_y}{s_x} \quad r = \frac{s_{xy}}{s_x s_y} \quad s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1} \quad b_0 = \bar{y} - b_1 \bar{x}$$

$$s_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

Intervalo de confianza de una predicción

Límite inferior	Límite superior
$L_i = \hat{y}_o - \left(t_{\frac{\alpha}{2}, n-2}\right) \cdot (s) \cdot \sqrt{1 + \frac{1}{n} + \frac{(x_o - \bar{x})^2}{S_{xx}}}$	$L_s = \hat{y}_o + \left(t_{\frac{\alpha}{2}, n-2}\right) \cdot (s) \cdot \sqrt{1 + \frac{1}{n} + \frac{(x_o - \bar{x})^2}{S_{xx}}}$

Siendo:

$$\hat{y}_o = (\hat{y}|x = x_o)$$

$$s = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}} = \sqrt{\frac{(1-r^2) \cdot s_y^2 \cdot (n-1)}{n-2}}$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$t_{\frac{\alpha}{2}, n-2} = \text{cuantil} \left(1 - \frac{\alpha}{2}\right) \text{ de T: Student}(n-2)$$

$$P\left\{T < t_{\frac{\alpha}{2}, n-2}\right\} = 1 - \frac{\alpha}{2} \rightarrow P\left\{T > t_{\frac{\alpha}{2}, n-2}\right\} = \frac{\alpha}{2} \quad (\text{Tabla A5})$$